

TARUNPREET SINGH PANESAR

LinkedIn: <https://www.linkedin.com/in/tarunpreet-singh-panesar/>
GitHub: <https://github.com/tarunpreet26>

Email: tarunpreet2439@gmail.com
Contact: +91-9877799838

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, Java, C, C++
- **Frameworks:** HTML5, CSS3
- **Tools & Platforms:** Git/GitHub, VS Code, AutoCAD, MySQL
- **Soft Skills:** Problem Prevention, Team Player, Adaptability, Multi-tasking

PROFESSIONAL TRAINING AND EXPERIENCE

- Data Analytics Essentials with AI: SQL, Python, Power BI, Tableau and Excel** | Centre of Professional Enhancement, Lovely Professional University (Location: Phagwara, Punjab) | [Certificate](#) Jun 2026 – Jul 2026
- Analyzed and transformed 55,500 healthcare records across 15 variables using Python (Pandas, NumPy) to derive demographic, clinical, billing, and admission insights.
 - Developed an interactive Power BI dashboard with 6 KPIs and 8 visualizations, analyzing \$1.4B+ in healthcare billing across ~40,000 hospital facilities.
 - Identified key trends across 6 medical conditions, 5 insurance providers, and 3 admission types, using interactive filters and cross-highlighting to enable data-driven analysis

PROJECTS

- Atmosfera** | HTML5, CSS3, Javascript, Python | [Github](#) May 2026
- Engineered a full-stack weather intelligence platform integrating real-time weather, 5-day forecasting, and AQI data through the OpenWeatherMap API.
 - Built an ensemble ML prediction pipeline using Random Forest, Gradient Boosting, and Linear Regression with 22 engineered atmospheric features for temperature forecasting.
 - Enhanced user accessibility across 6 languages through multilingual i18n and RTL support, complemented by interactive Chart.js visualizations and CSV, PDF, and Excel exports.
- Finova Advisor** | Python | [Github](#) May 2026
- Designed a 3-module AI-powered investment platform integrating Gemini 2.5 Flash, risk prediction personalized investment planning.
 - Deployed a decision tree classifier using 6 financial features to predict 3 risk levels, enabling personalized portfolio allocation and investment recommendations.
 - Optimized application performance with a 3-tier AI architecture, achieving chatbot responses within 1–3 seconds, investment plan generation in under 2 seconds, and 100% API error-handling coverage.
- Responsive Frontend for Real Estate Platform** | HTML5, CSS3, JavaScript | [Github](#) Nov 2024
- Developed a responsive and user-friendly frontend for a house rental platform, ensuring a clean and intuitive layout across devices.
 - Implemented JavaScript for dynamic interactions and integration with backend APIs, enabling smooth data flow between client and server.
 - Utilized HTML and CSS to structure and style the interface, delivering a consistent and visually appealing user experience.

RESEARCH PUBLICATIONS

- Energy-Efficient Semantic Communication Framework for Low-Power Wireless Body Area Network in XR Ecosystems** Aug 2026
- To be Presented:** Special Session 7- Future Internet Technologies: Internet of Everything (IoE), Web 4.0 and Intelligent Digital Ecosystems at International Conference on Transformative Technologies for Industry 5.0 to 6.0 (ICTTI-2026)
- To be Published:** Taylor & Francis

BOOK CHAPTERS

- Internet of Things and Wireless Communication: Principles and Architecture** Jul 2026
- Chapter Title:** Foundations of Machine Learning: Principles and Applications in IoT and Wireless Communication
- To be Published:** Taylor & Francis

CERTIFICATIONS

- **Database Management Systems** – [Infosys Springboard](#) Jul 2026
- **Object Oriented Programming** – [IamNeo](#) Jul 2025 – Jan 2026
- **Data Structures and Algorithms** – [IamNeo](#) Jul 2025 – Jan 2026
- **Programming using C++** – [Infosys Springboard](#) Sep 2025
- **Computer Programming using C language** – [IamNeo](#) Jan 2025 – May 2025

CURRICULAR ACTIVITIES

- **Community Member at 'Launch Crew', Initiative by Give My Certificate** Feb 2026 – Present
- **Member at 'Hack With India' LPU Chapter** Jul 2026 – Present

EDUCATION

- B.Tech – Computer Science & Engineering** | Lovely Professional University, Phagwara, Punjab Aug 2024 – Present
GPA: 7.58
- Senior Secondary (PCM)** | Doaba Public Senior Secondary School, Parowal, Punjab Apr 2022 – Mar 2024
GPA: 8.34
- Secondary Education (Matriculation)** | Mount Carmel School, Garhshankar, Punjab Apr 2020 – Mar 2022
GPA: 9.20